# Canonical Bundle Formula

We work over an algebraically closed field of characteristic zero, like $\mathbb{C}$. All varieties are assumed to be normal unless otherwise stated.

## 1 Birational divisors

**Definition 1** (b-divisors)**.** Let $X$ be a normal algebraic variety. We call $Y$ a *birational model over* $X$, or simply a *model over* $X$, if there exists a proper birational morphism $Y \to X$ of normal varieties. When both $X$ and $Y$ are quasi-projective, $Y \to X$ is projective.

Let $X' \dashrightarrow X$ be a birational map. For any prime divisor $E$ (resp. algebraic valuation $v$) over $X$, we denote by $c_{X'}(E)$ (resp. $v_{X'}$) the centre of $E$ (resp. $v$) on $X'$, whenever it exists. A *b-divisor* $\mathbf{D}$ on $X$ is a formal sum

$$\mathbf{D} = \sum_E r_E E \qquad \left(\text{resp. } \sum_v r_v v\right),$$

where the $E$ are prime divisors over $X$ (resp. the $v$ are algebraic valuations over $X$) and $r_E \in \mathbb{R}$ (resp. $r_v \in \mathbb{R}$), such that only finitely many terms with nonzero coefficient have a divisorial centre on $X$. If, in addition, $r_E \in \mathbb{Q}$ for every $E$ (resp. $r_v \in \mathbb{Q}$ for every $v$), then $\mathbf{D}$ is called a $\mathbb{Q}$-*b-divisor* on $X$.

The *trace* of $\mathbf{D}$ on a model $X' \to X$ is the $\mathbb{R}$-divisor

$$\operatorname{tr}_{X'} \mathbf{D} = D_{X'} = \sum_{E' \text{ a divisor on } X'} r_{E'} E' \qquad \left(\text{resp. } \sum_{v_{X'} \text{ a divisor}} r_v v_{X'}\right).$$

If $D_{X'}$ is $\mathbb{R}$-Cartier and $D_Y$ is the pullback of $D_{X'}$ on $Y$ for every model $Y \to X'$, we say that $\mathbf{D}$ *descends to* $X'$ and write

$$\mathbf{D} = \overline{D_{X'}}.$$

Let $X \to U$ be a proper morphism, and assume that $\mathbf{D}$ is a b-divisor over $X$ which descends to some model $Y \to X$. If $D_Y$ is nef/$U$ (resp. semi-ample/$U$), then we say that $\mathbf{D}$ is *b-nef/*$U$ (resp. *b-semi-ample/*$U$). If $D_Y$ is an $\mathbb{R}$-Cartier divisor, then we say that $\mathbf{D}$ is $\mathbb{R}$-*b-Cartier*. The terms b-Cartier and $\mathbb{Q}$-b-Cartier are defined similarly using Cartier and $\mathbb{Q}$-Cartier divisors, respectively. If $\mathbf{D}$ can be written as an $\mathbb{R}_{\geq 0}$-linear combination of nef/$U$ $\mathbb{Q}$-b-Cartier b-divisors, then we say that $\mathbf{D}$ is $NQC/U$.

When $U = \operatorname{Spec} k$, this gives the absolute setting.

**Definition 2** (Generalized pairs)**.** A *generalized sub-pair* (or *g-sub-pair* for short) is a triple $(X, B, \mathbf{M})/U$ consisting of a normal algebraic variety $X$ associated with a proper morphism $X \to U$, an $\mathbb{R}$-divisor $B$ on $X$, and an $\mathbb{R}$-b-Cartier b-nef/$U$ divisor $\mathbf{M}$ on $X$, such that $K_X + B + M_X$ is $\mathbb{R}$-Cartier. Note that, on a model $h : W \to X$ on which $\mathbf{M}$ descends, we put

$$M_X := h_* M_W.$$

If $B$ is a $\mathbb{Q}$-divisor and $\mathbf{M}$ is $\mathbb{Q}$-b-Cartier, then we say that $(X, B, \mathbf{M})/U$ is a $\mathbb{Q}$-*g-sub-pair*.

If $\mathbf{M} = 0 = \overline{0}$, a g-sub-pair $(X, B, \mathbf{M})/U$ is called a *sub-pair*. If $U = \{\text{pt}\}$, or if $U$ is not important, we drop $U$ from the notation.

A g-sub-pair (resp. $\mathbb{Q}$-g-sub-pair) $(X, B, \mathbf{M})/U$ is called a *g-pair* (resp. $\mathbb{Q}$-*g-pair*) if $B \geq 0$. A sub-pair $(X, B)$ is called a *pair* if $B \geq 0$.

**Definition 3** (Singularities of g-pairs)**.** Let $(X, B, \mathbf{M})/U$ be a g-sub-pair. For any prime divisor $E$ on $X$ and any $\mathbb{R}$-divisor $D$ on $X$, we define $\operatorname{mult}_E D$ to be the multiplicity of $D$ along $E$.

Let $h : W \to X$ be any log resolution of $(X, B)$ such that $\mathbf{M}$ descends to $W$, and write

$$K_W + B_W + M_W = h^*(K_X + B + M_X).$$

The *log discrepancy* of a prime divisor $E$ on $W$ with respect to $(X, B, \mathbf{M})$ is $1 - \operatorname{mult}_E B_W$, and it is denoted by $A(E, X, B, \mathbf{M})$. For the discrepancy

$$a(E, X, B, \mathbf{M}) = -\operatorname{mult}_E B_W,$$

we have

$$A(E, X, B, \mathbf{M}) = 1 + a(E, X, B, \mathbf{M}).$$

We say that $(X, B, \mathbf{M})$ is *sub-g-lc* (resp. *sub-g-klt*) if

$$A(E, X, B, \mathbf{M}) \geq 0 \qquad (\text{resp. } > 0)$$

for every log resolution $h : W \to X$ as above and every prime divisor $E$ on $W$.

Suppose that $(X, B, \mathbf{M})$ is sub-g-lc. A *g-lc place* of $(X, B, \mathbf{M})$ is a prime divisor $E$ over $X$ such that

$$A(E, X, B, \mathbf{M}) = 0.$$

A *g-lc centre* of $(X, B, \mathbf{M})$ is the centre of a g-lc place on $X$. We often say *lc place* and *lc centre* for short when the pair is clear.

We say that $(X, B, \mathbf{M})$ is *(sub-)g-dlt* if it is (sub-)g-lc and there exists an open subset $U \subseteq X$ such that

(1) $(U, B|_U)$ is log smooth;

(2) for any lc place $E$ over $X$, we have $c_X(E) \cap U \neq \varnothing$, and $c_X(E) \cap U$ is an lc centre of $(U, B|_U)$.

There is an important result in [Bir-Zhang14, proof of Lemma 4.5], also proved in [HL22, Proposition 3.9].

**Definition–Lemma 4** (G-dlt modifications)**.** Let $(X, B, \mathbf{M})/U$ be a quasi-projective g-lc g-pair. Then there is a projective birational morphism $r : X' \to X$ and a g-lc g-pair $(X', B', \mathbf{M})/U$ such that

(1) $(X', B', \mathbf{M})$ is $\mathbb{Q}$-factorial g-dlt;

(2) $K_{X'} + B' + M_{X'} = r^*(K_X + B + M_X)$; and

(3) any $r$-exceptional divisor is a component of $\lfloor B' \rfloor$.

For any projective birational morphism $r$ and $(X', B', \mathbf{M})/U$ which satisfy (1)–(3), $(r, (X', B', \mathbf{M})/U)$ will be called a *g-dlt modification* of $(X, B, \mathbf{M})$.

It has a useful application; see [JLX22, Lemma 2.6] or my note [Adj, 9, 10, 14, 15].

**Definition–Lemma 5** (G-dlt adjunction)**.** Let $(X, B, \mathbf{M})/U$ be a g-lc g-pair and let $S \subseteq \lfloor B \rfloor$ be a prime divisor, with normalization $\nu : S^\nu \to S$. Then there is a naturally defined g-pair structure $(S^\nu, B_{S^\nu}, \mathbf{M}^S)/U$ such that

$$K_{S^\nu} + B_{S^\nu} + M^S_{S^\nu} \sim_{\mathbb{R}} (K_X + B + M_X)|_{S^\nu}.$$

We say that $(S^\nu, B_{S^\nu}, \mathbf{M}^S)/U$ is the g-pair obtained by *divisorial adjunction*. If $(X, B, \mathbf{M})$ is g-lc, then $(S^\nu, B_{S^\nu}, \mathbf{M}^S)$ is g-lc, and if $(X, B, \mathbf{M})$ is g-dlt, then $(S^\nu, B_{S^\nu}, \mathbf{M}^S)$ is g-dlt.

For the construction of divisorial adjunction, see [Adj, 9 and 10]; for the g-dlt and g-lc parts, see [Adj, 14 and 15].

**Definition 6** (Discrepancy b-divisors)**.** Let $(X, B, \mathbf{M})/U$ be a g-sub-pair. For each model $f : Y \to X$, choose compatible canonical divisors with $f_*K_Y = K_X$, and set

$$\operatorname{tr}_Y \mathbf{A}(X, B, \mathbf{M}) = K_Y + M_Y - f^*(K_X + B + M_X).$$

These traces define the *discrepancy b-divisor* $\mathbf{A}(X, B, \mathbf{M})$; we often write its trace $A_Y$ for short. Define another b-divisor by

$$\operatorname{tr}_Y \mathbf{A}^*(X, B, \mathbf{M}) = A_Y^{>-1},$$

where we omit the components with coefficient at most $-1$. For a prime divisor $E$ over $X$,

$$\operatorname{mult}_E \mathbf{A}(X, B, \mathbf{M}) = A(E, X, B, \mathbf{M}) - 1 = a(E, X, B, \mathbf{M}),$$

where $a(E, X, B, \mathbf{M})$ is the discrepancy of $E$.

**Definition 7** (Sheaf associated to a b-divisor)**.** Let $\mathbf{D}$ be an $\mathbb{R}$-b-divisor on $X$. The sheaf $\mathcal{O}_X(\mathbf{D})$ is an $\mathcal{O}_X$-module whose sections on an open subset $U \subseteq X$ are given by

$$H^0\big(U, \mathcal{O}_X(\mathbf{D})\big) = \big\{\varphi \in k(X)^* \;\big|\; \operatorname{div}_Y(\varphi) + D_Y \geq 0 \text{ on } h^{-1}(U) \text{ for every model } h : Y \to X\big\} \cup \{0\},$$

where $k(X)$ is the function field of $X$. Note that $\mathcal{O}_X(\mathbf{D})$ is not necessarily coherent.

**Lemma 8.** *When* $\mathbf{D}$ *is b-Cartier,* $\mathcal{O}_X(\mathbf{D})$ *is coherent.*

*Proof.* Take a model $h : W \to X$ on which $\mathbf{D}$ descends. Then we have

$$\mathcal{O}_X(\mathbf{D}) = h_*\mathcal{O}_W(D_W).$$

Indeed, the inclusion $\mathcal{O}_X(\mathbf{D}) \subseteq h_*\mathcal{O}_W(D_W)$ is trivial. For the other inclusion, suppose that $\varphi \in H^0(h^{-1}(V), \mathcal{O}_W(D_W))$. This means that

$$\operatorname{div}_W(\varphi) + D_W \geq 0 \qquad \text{on } h^{-1}(V).$$

For any other model $Y \to X$, take a high common model

$$W \xleftarrow{\ r\ } T \xrightarrow{\ s\ } Y$$

over $X$. Since $\mathbf{D}$ descends to $W$,

$$\operatorname{div}_T(\varphi) + D_T = r^*\big(\operatorname{div}_W(\varphi) + D_W\big) \geq 0$$

on $(h \circ r)^{-1}(V)$. Pushing forward via $s$ gives

$$\operatorname{div}_Y(\varphi) + D_Y = s_*\big(\operatorname{div}_T(\varphi) + D_T\big) \geq 0$$

on $s((h \circ r)^{-1}(V))$. Thus $h_*\mathcal{O}_W(D_W) \subseteq \mathcal{O}_X(\mathbf{D})$, and coherence follows from the properness of $h$. □

For a b-divisor $\mathbf{D}$ on $X$, its *round up* $\lceil \mathbf{D} \rceil$ is defined by

$$\mathrm{tr}_Y \lceil \mathbf{D} \rceil = \lceil D_Y \rceil$$

on every model $Y \to X$. Similarly for round down.

**Definition 9** (Contraction)**.** A *contraction* is a proper surjective morphism $f : X \to Y$ of normal varieties with $f_* \mathcal{O}_X = \mathcal{O}_Y$.

Equivalently, $f$ has connected fibres. Indeed, in the Stein factorization

$$X \longrightarrow \widetilde{Y} \longrightarrow Y,$$

connected fibres force the finite morphism $\widetilde{Y} \to Y$ to have degree one, i.e., it is a birational morphism between normal varieties, so it is an isomorphism.

**Definition 10** (G-lc-trivial fibration)**.** Let $(X, B, \mathbf{M})/U$ be a g-sub-pair, and let $f : X \to Y$ be a contraction over $U$. We call $f : (X, B, \mathbf{M}) \to Y$ a *g-lc-trivial fibration* over $U$ if

(1) $(X, B, \mathbf{M})$ is g-sub-lc over the generic point of $Y$;

(2) $\operatorname{rank} f_* \mathcal{O}_X \big( \lceil \mathbf{A}^*(X, B, \mathbf{M}) \rceil \big) = 1$; and

(3) there exists an $\mathbb{R}$-Cartier divisor $L$ on $Y$ such that

$$K_X + B + M_X \sim_{\mathbb{R}} f^* L.$$

If, in addition, $(X, B, \mathbf{M})$ is g-sub-klt over the generic point of $Y$, we say *g-klt-trivial*. For $\mathbf{M} = 0$, these are classical lc-trivial and klt-trivial fibrations; see [FG14, Definitions 3.1–3.2].

*Remark* 11. Condition (3) is sometimes written as

$$K_X + B + M_X \sim_{\mathbb{R}, Y} 0.$$

When $f$ is a contraction, they are equivalent.

**Lemma 12** ([Fuj12, Lemmas 3.19–3.20])**.** *Let $(X, B, \mathbf{M})/U$ be a g-sub-lc g-sub-pair, and let $h : W \to X$ be a log resolution on which $\mathbf{M}$ descends. Write*

$$K_W + B_W + M_W = h^*(K_X + B + M_X).$$

*Then*

$$\mathcal{O}_X \big( \lceil \mathbf{A}^*(X, B, \mathbf{M}) \rceil \big) = h_* \mathcal{O}_W \big( \lceil -B_W^{<1} \rceil \big).$$

*In particular, this sheaf is coherent. If $B \geq 0$, then*

$$\mathcal{O}_X \big( \lceil \mathbf{A}^*(X, B, \mathbf{M}) \rceil \big) = \mathcal{O}_X.$$

*Proof.* By definition,

$$\mathrm{tr}_W \, \mathbf{A}(X, B, \mathbf{M}) = -B_W, \qquad \mathrm{tr}_W \, \mathbf{A}^*(X, B, \mathbf{M}) = -B_W^{<1}.$$

Let $g : W' \to W$ be any model over $W$. Define $B_{W'}$ by

$$K_{W'} + B_{W'} + M_{W'} = (h \circ g)^*(K_X + B + M_X).$$

Since $\mathbf{M}$ descends to $W$, we have $M_{W'} = g^* M_W$. Cancelling these terms gives

$$K_{W'} + B_{W'} = g^*(K_W + B_W).$$

*Step 1.* We claim that

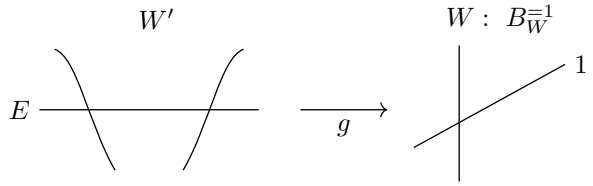
$$\lceil \mathbf{A}^*(X, B, \mathbf{M})_{W'} \rceil = g^* \lceil \mathbf{A}^*(X, B, \mathbf{M})_W \rceil + F,$$

where $F \geq 0$ is an integral $g$-exceptional divisor.

The divisor $\lceil -B_W^{<1} \rceil$ is an effective Cartier divisor, and is supported on the negative components of $B_W$. Thus the pair

$$\big(W, B_W + \lceil -B_W^{<1} \rceil\big)$$

is log smooth and has lc singularities, since for a coefficient $b < 1$, we have $0 \leq b + \lceil -b \rceil < 1$. It also has the same lc places as $(W, B_W)$, because an lc place of either log smooth pair has centre a stratum of $B_W^{=1}$, but $\operatorname{Supp} \lceil -B_W^{<1} \rceil$ contains no such stratum.




$$a(E) = 2 - 1 - (1 + 1) = -1$$

For a prime divisor $E$ on $W'$, the number $\operatorname{mult}_E g^* \lceil -B_W^{<1} \rceil$ is a nonnegative integer, since $g^* \lceil -B_W^{<1} \rceil \geq 0$. If $\operatorname{mult}_E B_{W'} = 1$, then $E$ is an lc place of $(W, B_W)$, so its multiplicity in $g^* \lceil -B_W^{<1} \rceil$ is zero. Both sides of the claimed equality

$$\lceil -B_{W'}^{<1} \rceil = g^* \lceil -B_W^{<1} \rceil + F$$

then give $\operatorname{mult}_E F = 0$.

If $\operatorname{mult}_E B_{W'} < 1$, we have

$$A(E, W, B_W) = 1 - \operatorname{mult}_E B_{W'} > 0,$$

so $E$ is not an lc place of $(W, B_W)$. Since the two pairs have the same lc places, we have

$$\begin{aligned} 0 &< A\big(E, W, B_W + \lceil -B_W^{<1} \rceil\big) \\ &= 1 - \operatorname{mult}_E B_{W'} - \operatorname{mult}_E g^* \lceil -B_W^{<1} \rceil. \end{aligned}$$

Rounding this inequality gives

$$\lceil -\operatorname{mult}_E B_{W'} \rceil \geq \operatorname{mult}_E g^* \lceil -B_W^{<1} \rceil \geq 0.$$

The left-hand side is nothing but $\operatorname{mult}_E \lceil -B_{W'}^{<1} \rceil$, so $\operatorname{mult}_E F \geq 0$. Since $(W, B_W)$ is sub-lc, the coefficients of $B_{W'}$ are at most 1. Therefore $F \geq 0$ is effective. Since both sides of the claimed equality are integral divisors, $F$ is integral.

Finally, if $E$ is not $g$-exceptional, then $g(E)$ is a divisor on $W$. That is, $g$ is an isomorphism over the generic point of $g(E)$ by normality, so we have

$$\operatorname{mult}_E B_{W'} = \operatorname{mult}_{g(E)} B_W.$$

Then

$$\operatorname{mult}_E F = \operatorname{mult}_E \lceil -B^{<1}_{W'} \rceil - \operatorname{mult}_E g^* \lceil -B^{<1}_{W} \rceil = 0.$$

We see that $F$ is $g$-exceptional. This proves the claim.

*Step 2.* By algebraic Hartogs, we have

$$g_* \mathcal{O}_{W'}(F) = \mathcal{O}_W.$$

Then we have

$$\begin{aligned} g_* \mathcal{O}_{W'}\big(\lceil \mathbf{A}^*(X, B, \mathbf{M})_{W'} \rceil\big) &= g_* \mathcal{O}_{W'}\big(g^* \lceil -B^{<1}_W \rceil + F\big) \\ &\cong \mathcal{O}_W\big(\lceil -B^{<1}_W \rceil\big), \end{aligned}$$

thanks to the projection formula. These identifications are induced by the identity on rational functions.

For any other model $\psi : Y \to X$, take a common higher model

$$W \xleftarrow{r} T \xrightarrow{s} Y.$$

Fix any open subset $V \subseteq X$. Suppose that we have

$$\operatorname{div}_W(\varphi) + \lceil -B^{<1}_W \rceil \geq 0$$

on $h^{-1}(V)$. Applying the claimed equality to $r : T \to W$, we have

$$r_* \mathcal{O}_T\big(\lceil \mathbf{A}^*(X, B, \mathbf{M})_T \rceil\big) \cong \mathcal{O}_W\big(\lceil -B^{<1}_W \rceil\big).$$

Thus it gives

$$\operatorname{div}_T(\varphi) + \lceil \mathbf{A}^*(X, B, \mathbf{M})_T \rceil \geq 0$$

on $(h \circ r)^{-1}(V)$. We push forward this along $s : T \to Y$ and get

$$\operatorname{div}_Y(\varphi) + \lceil \mathbf{A}^*(X, B, \mathbf{M})_Y \rceil = s_*\big(\operatorname{div}_T(\varphi) + \lceil \mathbf{A}^*(X, B, \mathbf{M})_T \rceil\big) \geq 0$$

on $s((h \circ r)^{-1}(V)) = \psi^{-1}(V)$. It gives

$$h_* \mathcal{O}_W\big(\lceil -B^{<1}_W \rceil\big) \subseteq \mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M}) \rceil\big).$$

The reverse direction is trivial. Coherence is due to the properness of $h$.

Finally, if $B \geq 0$, then every non-exceptional component of $B_W$ has coefficient in $[0, 1]$. Hence $\lceil -B^{<1}_W \rceil$ is effective and $h$-exceptional. By algebraic Hartogs, we have

$$h_* \mathcal{O}_W\big(\lceil -B^{<1}_W \rceil\big) = \mathcal{O}_X.$$

□

Now let $f : X \to Y$ be a contraction over $U$. If $(X, B, \mathbf{M})$ is only g-sub-lc over the generic point of $Y$, i.e., it is g-sub-lc over an open subset $Y^0 \subseteq Y$, applying Lemma 12, we have

$$\mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M}) \rceil\big) = h_* \mathcal{O}_W\big(\lceil -B^{<1}_W \rceil\big)$$

on $f^{-1}(Y^0)$. Pushing forward along $f$, we get

$$f_* \mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M}) \rceil\big) = (f \circ h)_* \mathcal{O}_W\big(\lceil -B^{<1}_W \rceil\big)$$

on $Y^0$, and since $\eta_Y \in Y^0$, we have

$$\operatorname{rank} f_* \mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M}) \rceil\big) = \operatorname{rank} (f \circ h)_* \mathcal{O}_W\big(\lceil -B^{<1}_W \rceil\big).$$

**Lemma 13** (The rank-one condition)**.** *Let $(X, B, \mathbf{M})/U$ be a g-sub-pair, and let $f : X \to Y$ be a contraction over $U$. Suppose that $(X, B, \mathbf{M})$ is g-sub-lc over the generic point of $Y$ and that $B$ is effective over that point, i.e., its components which dominate $Y$ have nonnegative coefficients. Then*

$$\operatorname{rank} f_*\mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M})\rceil\big) = 1.$$

*Proof.* We may replace $Y$ by an open subset and assume that $B \geq 0$ and $(X, B, \mathbf{M})$ is g-lc; this does not change the rank. Take $h : W \to X$ to be a log resolution on which $\mathbf{M}$ descends. The divisor $\lceil -B^{<1}_W\rceil$ is effective and $h$-exceptional. So by Lemma 12,

$$\operatorname{rank} f_*\mathcal{O}_X\big(\lceil \mathbf{A}^*(X, B, \mathbf{M})\rceil\big) = \operatorname{rank} f_*\mathcal{O}_X = \operatorname{rank} \mathcal{O}_Y = 1.$$

□

**Example 14** ($B \not\geq 0$ over the generic point)**.** Put $f : \mathbb{P}^1 \to \operatorname{Spec}\mathbb{C}$, $B = -p$, and $\mathbf{M} = 0$, where $p \in \mathbb{P}^1$. The pair $(\mathbb{P}^1, B)$ is a sub-klt pair. Take $h = \mathrm{id}$. Then

$$h_*\mathcal{O}_{\mathbb{P}^1}\big(\lceil -B^{<1}\rceil\big) = \mathcal{O}_{\mathbb{P}^1}(p).$$

Therefore,

$$\operatorname{rank} f_*\mathcal{O}_{\mathbb{P}^1}(p) = \dim_{\mathbb{C}} H^0\big(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^1}(p)\big) = 2,$$

not rank one.

# 2 Divisorial Thresholds: Discriminant and Moduli b-divisor

For a contraction $f : (X, B, \mathbf{M}) \to Y$ where $(X, B, \mathbf{M})$ is g-sub-lc over the generic point of $Y$, we fix an $\mathbb{R}$-Cartier divisor $L$ on $Y$ and an $\mathbb{R}$-linear equivalence

$$K_X + B + M_X \sim_{\mathbb{R}} f^*L.$$

Let $P$ be a prime divisor over $Y$. Choose a smooth model $\psi : Y' \to Y$ on which $P$ appears. Resolve the main component of $X \times_Y Y'$ and take a sufficiently high model $W$ on which $\mathbf{M}$ descends. We obtain the commutative diagram

$$\begin{array}{ccc} W & \xrightarrow{h} & X \\ {\scriptstyle f_W}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow[\psi]{} & Y. \end{array}$$

We put

$$K_W + B_W + M_W = h^*(K_X + B + M_X).$$

**Definition 15.** The *generalized lc threshold* over $\eta_P$, the generic point of $P$, is

$$t_P = \sup\big\{s \in \mathbb{R} \mid (W, B_W + sf^*_W P, \mathbf{M}) \text{ is g-sub-lc over } \eta_P\big\}.$$

The condition is tested over an open neighborhood of $\eta_P$ in $Y'$, and the number $s$ may be negative.

**Lemma 16** (Threshold formula)**.** *After replacing $W$ by a higher log resolution, we may assume that* $\operatorname{Supp} B_W \cup \operatorname{Supp} f_W^* P$ *is snc over $\eta_P$. Let $E_1, \dots, E_n$ be the components of this union meeting the fibre $f_W^{-1}(\eta_P)$, and put*

$$b_i = \operatorname{mult}_{E_i} B_W, \qquad m_i = \operatorname{mult}_{E_i} f_W^* P.$$

*Then*

$$t_P = \min_{m_i > 0} \frac{1 - b_i}{m_i}. \tag{2.1}$$

*The value of $t_P$ is independent of the choice of log resolution.*

*Proof.* Since $\mathbf{M}$ descends to $W$, the generalized discrepancies coincide with the discrepancies of $(W, B_W + s f_W^* P)$. The snc condition gives that $(W, B_W + s f_W^* P)$ is sub-lc over $\eta_P$ if and only if $b_i + s m_i \le 1$ for every $i$, see [BGR, Lemma 5.2.20]. Since $f_W$ is proper and $E_i$ meets the fibre over $\eta_P$, its image is either $P$ or $Y'$. If $m_i = 0$, such a component $E_i$ will dominates $Y'$, and $b_i \le 1$ by sub-lc over $\eta_P$. If $m_i > 0$, the condition is $s \le (1 - b_i)/m_i$. At least one such component exists, since $f_W$ is surjective. This proves (2.1).

On a common higher resolution over $Y'$, the crepant pullbacks preserves the discrepancies, hence $t_P$ is independent of the resolution. See [BGR, Lemma 5.2.16]. □

**Definition 17** (Discriminant and moduli part)**.** For each prime divisor $P$ over $Y$, put $\operatorname{mult}_P \mathbf{C} = 1 - t_P$. On a smooth model $\psi : Y' \to Y$, define

$$C_{Y'} = \sum_{P \subset Y'} (1 - t_P) P, \qquad N_{Y'} = \psi^* L - K_{Y'} - C_{Y'},$$

where the sum runs over the prime divisors on $Y'$. The b-divisor $\mathbf{C}$ is called the *discriminant*, and $\mathbf{N} = \overline{L} - \mathbf{K} - \mathbf{C}$ is the *moduli b-divisor*, where $\mathbf{K}$ has traces given by compatible canonical divisors on the models of $Y$.

On other models their traces are obtained by pushforward from a dominating smooth model.

**Proposition 18.** $\mathbf{C}$ *and* $\mathbf{N}$ *are b-divisors, and on every model $\psi : Y' \to Y$ their traces satisfy*

$$K_{Y'} + C_{Y'} + N_{Y'} = \psi^* L.$$

*Proof. Step 1.* Fix a smooth model $\psi : Y' \to Y$. Choose a log resolution $h : W \to X$ mapping to $Y'$ on which $\mathbf{M}$ descends and $\operatorname{Supp} B_W$ is snc. We use the same diagram

$$\begin{array}{ccc} W & \xrightarrow{h} & X \\ {\scriptstyle f_W}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow[\psi]{} & Y. \end{array}$$

By generic smoothness, there is a nonempty open subset $U \subseteq Y'$ such that $f_W^{-1}(U) \to U$ is smooth. By properness of $f_W$, we may shrink $U$ further to cut off the vertical components of $B_W$. After shrinking $U$ further, the remaining horizontal components have relative snc support over $U$. Their intersections are smooth over $U$, and their restrictions to every (possibly non-closed) fibre have snc support.

For a prime divisor $P$ on $Y'$ meeting $U$, its generic point $\eta_P$ lies in $U$. Take an open subset $\eta_P \in V \subseteq U$ such that $P_V = P \cap V$ is smooth, and set $W_V = f_W^{-1}(V)$. The pullback

divisor over $V$ has the following scheme-theoretic description.[1]

$$D := f_W^* P|_{W_V} \cong (P \times_{Y'} W) \times_W (V \times_{Y'} W) \cong W_V \times_V P_V.$$

Since $W_V \to V$ is smooth, $D \to P_V$ is smooth. As $P_V$ is smooth, $D$ is smooth and hence reduced. In particular, $f_W^* P$ is reduced over $\eta_P$.

At any point of $D$, let $H_1, \dots, H_r$ be the components of $\operatorname{Supp} B_W$ passing through that point, and set $S = \bigcap_{i=1}^r H_i$ (with $S = W$ if $r = 0$). Write $S_V = S \cap W_V$. We have

$$S \cap D \cong S \times_W (W_V \times_V P_V) \cong S \times_{Y'} P_V \cong S_V \times_V P_V.$$

This is the scheme-theoretic inverse image of $P_V$ under $f_W|_{S_V} : S_V \to V$. Since $S \cap f_W^{-1}(U) \to U$ is smooth by the relative snc condition, $S_V \to V$ is smooth. Thus $S \cap D$ is smooth and, where nonempty, is an effective Cartier divisor in $S_V$, by smooth base change of the smooth Cartier divisor $P_V$. Therefore $D$ meets the horizontal boundary strata transversely.

On $W_V$ write

$$B_W|_{W_V} = \sum_i b_i H_i, \qquad D = \sum_j D_j,$$

where $D$ is reduced and the $H_i$ are the horizontal components restricted to $W_V$. Each $D_j$ maps into $P_V$, so no $D_j$ is a component of $B_W|_{W_V}$. Then

$$B_W|_{W_V} + sD = \sum_i b_i H_i + \sum_j s D_j,$$

where every $b_i \le 1$ because $(X, B, \mathbf{M})$ is g-sub-lc over the generic point of $Y$. Since $\mathbf{M}$ descends to $W$, the g-sub-pair $(W, B_W + s f_W^* P, \mathbf{M})$ is g-sub-lc over $\eta_P$ if and only if the sub-pair $(W, B_W + s f_W^* P)$ is sub-lc over $\eta_P$. This sub-pair is log smooth over $\eta_P$ by the transversality proved above. From the displayed coefficients, it is sub-lc over $\eta_P$ if and only if $s \le 1$. Thus $t_P = 1$. Only the finitely many prime divisors in $Y' \setminus U$ can contribute to $C_{Y'}$.

*Step 2.* Let $r : Y'' \to Y'$ be a higher smooth model. If $\widetilde{P}$ is the strict transform of $P$, they define the same divisorial valuation. Since $r$ is an isomorphism over a neighbourhood of $\eta_P$, the thresholds agree: $t_{\widetilde{P}} = t_P$. The $r$-exceptional terms push forward to zero, so

$$r_* C_{Y''} = \sum_{P \subset Y'} (1 - t_{\widetilde{P}}) r_* \widetilde{P} = \sum_{P \subset Y'} (1 - t_P) P = C_{Y'}.$$

Choose compatible canonical divisors, i.e., traces of $\mathbf{K}$. Then

$$r_* N_{Y''} = r_* \big( r^* \psi^* L - K_{Y''} - C_{Y''} \big) = \psi^* L - K_{Y'} - C_{Y'} = N_{Y'}.$$

A common higher smooth model therefore makes the pushforwards to any normal model independent of the smooth model used to compute them. This proves that $\mathbf{C}$ and $\mathbf{N}$ are b-divisors. Pushing forward the identity on a smooth model also gives the stated identity on every model. □

[1] Here we use the scheme-theoretic notation $P_V = P \cap V \cong P \times_{Y'} V$ and $W_V = f_W^{-1}(V) \cong V \times_{Y'} W$. The displayed identification follows by transitivity of fibre products [Görtz I, Proposition 4.16].

Note that if

$$L' = L + \sum_j r_j \operatorname{div}_Y(u_j), \qquad r_j \in \mathbb{R}, \quad u_j \in k(Y)^\times,$$

then the same discriminant $\mathbf{C}$ occurs for $L'$ and

$$N'_{Y'} = N_{Y'} + \sum_j r_j \operatorname{div}_{Y'}(u_j)$$

on each model $\psi : Y' \to Y$, where the $u_j$ are viewed as rational functions on $Y'$. Thus changing $L$ by an $\mathbb{R}$-principal divisor changes $\mathbf{N}$ by the corresponding $\mathbb{R}$-principal b-divisor.

**Theorem 19** (Canonical bundle formula)**.** *Let $f : X \to Y$ be a contraction of normal quasi-projective varieties over $U$. Let $(X, B, \mathbf{M})/U$ be a g-pair which is g-lc over the generic point of $Y$, and suppose that*

$$K_X + B + M_X \sim_{\mathbb{R}} f^*L,$$

*where $L$ is an $\mathbb{R}$-Cartier divisor on $Y$. Then the moduli b-divisor $\mathbf{N}$ is b-nef/$U$ and $\mathbb{R}$-b-Cartier.*

*Proof.* By Lemma 13, $f : (X, B, \mathbf{M}) \to Y$ is a g-lc-trivial fibration. Applying [CHLX23, Theorem 11.4.4] gives the result. □

Hashizume gives an alternative proof in [Has26, Theorem 1.3]. This theorem requires no NQC condition.

**Proposition 20.** *Under the hypotheses of Theorem 19, we have $C_Y \geq 0$ and $(Y, C_Y, \mathbf{N})/U$ is a g-pair satisfying*

$$K_Y + C_Y + N_Y = L,$$

*where $N_Y = \operatorname{tr}_Y \mathbf{N}$. For every prime divisor $P$ over $Y$,*

$$A(P, Y, C_Y, \mathbf{N}) = t_P.$$

*Moreover, if $(X, B, \mathbf{M})$ is a g-lc (resp. g-klt) g-pair, then $(Y, C_Y, \mathbf{N})$ is a g-lc (resp. g-klt) g-pair.*

*Proof. Step 1.* First we prove that $C_Y \geq 0$. Let $P$ be a prime divisor on $Y$. Since $Y$ is normal, we may choose an open subset $V \subseteq Y$ containing $\eta_P$ on which $P|_V$ is Cartier. After shrinking $V$, write

$$P|_V = \operatorname{div}_V(\varphi)$$

as an effective Cartier divisor. Put $V' = f^{-1}(V)$ and $f_V = f|_{V'} : V' \to V$. Its pullback is

$$f_V^*(P|_V) = \operatorname{div}_{V'}(\varphi \circ f).$$

Since $f$ is proper and surjective, this divisor has a prime component dominating $P|_V$. Let $E \subseteq X$ be its closure in $X$, and put

$$m = \operatorname{mult}_E f_V^*(P|_V) > 0, \qquad b = \operatorname{mult}_E B \geq 0.$$

Adding $s f_V^*(P|_V)$ to $B|_{V'}$ gives multiplicity $b + sm$ along $E$. Thus g-sub-lc over $\eta_P$ requires $b + sm \leq 1$. We may compute the threshold on a sufficiently high model on which $\mathbf{M}$ descends and the crepant pullback has snc support over $\eta_P$. Consequently,

$$t_P \leq \frac{1-b}{m} \leq 1, \qquad \operatorname{mult}_P C_Y = 1 - t_P \geq 0,$$

where $m$ is a positive integer. This holds for every prime divisor $P$ on $Y$, so $C_Y \geq 0$.

*Step 2.* By definition, $N_Y = L - K_Y - C_Y$. The divisor $L$ is $\mathbb{R}$-Cartier, and Theorem 19 gives that $\mathbf{N}$ is b-nef/$U$ and $\mathbb{R}$-b-Cartier. Hence $(Y, C_Y, \mathbf{N})/U$ is a g-pair.

Let $P$ be any prime divisor over $Y$. Choose a smooth model $\psi : Y' \to Y$ on which $P$ appears and $\mathbf{N}$ descends. The definition of $N_{Y'}$ gives

$$K_{Y'} + C_{Y'} + N_{Y'} = \psi^* L = \psi^*(K_Y + C_Y + N_Y).$$

Thus $C_{Y'}$ is the crepant boundary used to compute the log discrepancies of $(Y, C_Y, \mathbf{N})$. Since $\operatorname{mult}_P C_{Y'} = 1 - t_P$, we have

$$A(P, Y, C_Y, \mathbf{N}) = 1 - \operatorname{mult}_P C_{Y'} = t_P.$$

*Step 3.* It remains to prove the singularity assertions. Suppose that $(X, B, \mathbf{M})$ is g-lc. For the prime divisor $P \subseteq Y'$ chosen in Step 2, take a log resolution $h : W \to X$ with a commutative diagram

$$\begin{array}{ccc} W & \xrightarrow{g} & Y' \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle \psi} \\ X & \xrightarrow[f]{} & Y \end{array}$$

on which $\mathbf{M}$ descends and $\operatorname{Supp} B_W \cup \operatorname{Supp} g^* P$ has snc support. Let $E_1, \ldots, E_n$ be the components of $g^* P$ dominating $P$, and set

$$m_i = \operatorname{mult}_{E_i} g^* P > 0, \qquad b_i = \operatorname{mult}_{E_i} B_W \leq 1.$$

Since $\mathbf{M}$ descends to $W$, the g-sub-pair $(W, B_W + s g^* P, \mathbf{M})$ is g-sub-lc over $\eta_P$ if and only if the sub-pair $(W, B_W + s g^* P)$ is sub-lc over $\eta_P$. By the snc condition, this is equivalent to $b_i + s m_i \leq 1$ for every $i$; see [BGR, Corollary 5.2.20]. Lemma 16 gives

$$t_P = \min_{m_i > 0} \frac{1 - b_i}{m_i} \geq 0.$$

Thus $(Y, C_Y, \mathbf{N})$ is g-lc. If $(X, B, \mathbf{M})$ is g-klt, every $b_i < 1$, so the same formula gives $t_P > 0$. Hence $(Y, C_Y, \mathbf{N})$ is g-klt. □

**Example 21.** Let $E$ be an elliptic curve, let $Y$ be a smooth projective variety, and let $P \subseteq Y$ be a smooth prime divisor. Set $f : E \times Y \to Y$, $B = b f^* P$, and $\mathbf{M} = 0$, where $b \geq 0$. Choose a nowhere-vanishing differential form on $E$, so its canonical divisor is zero. Then

$$K_{E \times Y} + B \sim_{\mathbb{R}} f^*(K_Y + bP).$$

The pair $(E \times Y, B)$ is klt over the generic point of $Y$ because the boundary is vertical. Since $f^* P = E \times P$ is smooth and reduced, and

$$B + s f^* P = (b + s)(E \times P),$$

Lemma 16 gives $t_P = 1 - b$.

For every prime divisor $Q \subseteq Y$ different from $P$, the boundary $B$ has empty support over $\eta_Q$, so $t_Q = 1$. Taking $L = K_Y + bP$, we obtain

$$C_Y = bP, \qquad N_Y = L - K_Y - C_Y = 0.$$

For example, if $b = 2$, the pair $(Y, 2P)$ is not lc, although $(E \times Y, 2 f^* P)$ is klt over the generic point of $Y$, and $t_P = -1$.

**Lemma 22** (Restriction of a divisorial valuation)**.** *Let $f : X \to Y$ be a dominant morphism of algebraic varieties. Let $E$ be a prime divisor over $X$, and suppose that $\operatorname{ord}_E|_{k(Y)}$ is nontrivial. Then there are a model $\psi : Y' \to Y$, a prime divisor $P \subseteq Y'$, and an integer $e > 0$ such that*

$$\operatorname{ord}_E|_{k(Y)} = e \operatorname{ord}_P .$$

*Here the restriction is the valuation*

$$v : k(Y)^\times \longrightarrow \mathbb{Z}, \qquad u \longmapsto \operatorname{ord}_E\big((f \circ h)^* u\big),$$

*where $h : W \to X$ is a model carrying $E$.*

*Proof.* Consider the valuation rings

$$R_E = \mathcal{O}_{W,\eta_E}, \qquad R_v = R_E \cap k(Y),$$

where we identify $k(Y) \subseteq k(X) = k(W)$ using the dominant morphism $f$ and the birational morphism $h$. Denote their residue fields by

$$k(E) = R_E/\mathfrak{m}_E, \qquad k(v) = R_v/\mathfrak{m}_v.$$

The value group of $v$ is a nonzero subgroup of $\mathbb{Z}$, so $R_v$ is a discrete valuation ring.

*Step 1.* We first prove

$$s := \operatorname{trdeg}_{k(v)} k(E) \le \operatorname{trdeg}_{k(Y)} k(X) = \dim X - \dim Y.$$

Lift algebraically independent elements $\overline{u}_1, \dots, \overline{u}_s$ of $k(E)$ over $k(v)$ to $u_1, \dots, u_s \in R_E$. These lifts have value zero, since they lie outside $\mathfrak{m}_E$, and are algebraically independent over $k(Y)$. This proves the inequality.

Since $E \subseteq W$ is a prime divisor, we have

$$\begin{aligned} d := \operatorname{trdeg}_k k(v) &= \operatorname{trdeg}_k k(E) - \operatorname{trdeg}_{k(v)} k(E) \\ &= \dim X - 1 - s \ge \dim Y - 1. \end{aligned}$$

For the reverse inequality, lift a transcendence basis $\overline{z}_1, \dots, \overline{z}_d$ of $k(v)/k$ to $z_1, \dots, z_d \in R_v$. Each lift has value zero. Choose $t \in k(Y)$ with $v(t) > 0$. For a polynomial

$$\sum_i a_i t^i, \qquad a_i \in k[z_1, \dots, z_d] \setminus \{0\},$$

we have

$$v(a_i t^i) = v(a_i) + i\, v(t) = i\, v(t).$$

Let $i_0$ be the least exponent appearing in the sum. We write

$$\sum_i a_i t^i = t^{i_0} \left( a_{i_0} + \sum_{i > i_0} a_i t^{i - i_0} \right),$$

where

$$\overline{a_{i_0} + \sum_{i > i_0} a_i t^{i-i_0}} = \overline{a}_{i_0} \ne 0 \qquad \text{in } k(v).$$

Thus the term in parentheses is nonzero, which implies that $\sum_i a_i t^i \ne 0$. Hence $z_1, \dots, z_d, t$ are algebraically independent over $k$, i.e., $d + 1 \le \dim Y$. So

$$\operatorname{trdeg}_k k(v) = \dim Y - 1.$$

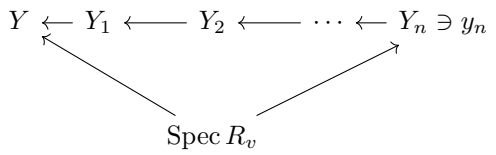
*Step 2.* Let $x \in X$ be the centre of $R_E$ on $X$, i.e., the generic point of $c_X(E)$, and put $y = f(x)$. The inclusion

$$\mathcal{O}_{Y,y} \longrightarrow R_E \cap k(Y) = R_v$$

is a local homomorphism, so $R_v$ has centre $y$ on $Y$. By [BGR, Lemma 5.2.13], a finite sequence of blow-ups of its successive centres gives a proper birational morphism $Y_n \to Y$ and a point $y_n \in Y_n$ such that

$$\mathcal{O}_{Y_n,y_n} \cong R_v.$$

The corresponding morphisms fit in the diagram




where $\mathfrak{m}_v \in \operatorname{Spec} R_v$ maps to the successive centres, and

$$\operatorname{trdeg}_k k(y_n) = \operatorname{trdeg}_k k(v) = \dim Y - 1.$$

*Step 3.* Take the normalization $Y' \to Y_n$. Since $R_v$ is a DVR and hence normal, the local ring at the point above $y_n$ is still $R_v$. This is a codimension-one point, whose closure is a prime divisor $P \subseteq Y'$ satisfying

$$\mathcal{O}_{Y',\eta_P} \cong \mathcal{O}_{Y_n,y_n} \cong R_v.$$

Let $e$ be the value under $v$ of a uniformizer of $\mathcal{O}_{Y',\eta_P}$. Then $e > 0$ and $v = e \operatorname{ord}_P$, as required. □

**Proposition 23** (Images of lc centres)**.** *Assume that $(X, B, \mathbf{M})$ in Theorem 19 is a g-lc g-pair. Then the lc centres of $(Y, C_Y, \mathbf{N})$ are exactly the images of the vertical lc centres of $(X, B, \mathbf{M})$.*

*Proof.* Let $P$ be an lc place of $(Y, C_Y, \mathbf{N})$. By Proposition 20, $t_P = 0$. Lemma 16 gives that this minimum is attained by a component $E$ with

$$\operatorname{mult}_E f_W^* P > 0, \qquad \operatorname{mult}_E B_W = 1$$

on a log resolution $h : W \to X$ mapping to a model $Y'$ carrying $P$, on which $\mathbf{M}$ descends. We have the diagram

$$\begin{array}{ccc} W & \xrightarrow{f_W} & Y' \supseteq P \\ {\scriptstyle h}\downarrow & & \downarrow{\scriptstyle \psi} \\ X & \xrightarrow[f]{} & Y. \end{array}$$

Then $E$ is an lc place of $(X, B, \mathbf{M})$ and dominates $P$. Its centre $c_X(E)$ maps onto $c_Y(P)$.

Conversely, let $E$ be an lc place of $(X, B, \mathbf{M})$ whose centre on $X$ does not dominate $Y$. Put

$$Z = f(c_X(E)) \subsetneq Y.$$

Let $x \in X$ be the generic point of $c_X(E)$ and $y = f(x)$. Choose a nonzero $u \in \mathfrak{m}_{Y,y}$, viewed as a rational function regular near $y$ and vanishing along $Z$. The local homomorphisms

$$\mathcal{O}_{Y,y} \longrightarrow \mathcal{O}_{X,x} \longrightarrow R_E$$

send $u$ into $\mathfrak{m}_E$, so $\operatorname{ord}_E((f\circ h)^*u) > 0$. Thus $\operatorname{ord}_E|_{k(Y)}$ is nontrivial. By Lemma 22, it is $e\operatorname{ord}_P$ for a prime divisor $P$ over $Y$.

Take a smooth model $\psi : Y' \to Y$ carrying $P$, and a log resolution $h : W \to X$ carrying $E$, on which $\mathbf{M}$ descends, with a morphism $f_W : W \to Y'$. Identify $k(Y) = k(Y')$ and put $y' = f_W(\eta_E)$. There is a local homomorphism

$$\mathcal{O}_{Y',y'} \longrightarrow R_E \cap k(Y') = R_v,$$

so $y'$ is a centre of $v$. Since $R_v \cong \mathcal{O}_{Y',\eta_P}$, the point $\eta_P$ is also its centre. As $Y'$ is separated, the centre is unique. Therefore $f_W(E) = P$. Moreover, if $u$ is a local equation of $P$ near $\eta_P$, with $\operatorname{ord}_P(u) = 1$, then

$$\begin{aligned}\operatorname{mult}_E f_W^* P &= \operatorname{ord}_E(f_W^* u) = v(u)\\ &= e\operatorname{ord}_P(u) = e > 0.\end{aligned}$$

Since $E$ is an lc place, $A(E, X, B, \mathbf{M}) = 0$, and hence

$$\operatorname{mult}_E(B_W + sf_W^* P) = 1 + se.$$

Thus g-sub-lc over $\eta_P$ requires $1 + se \le 1$, i.e., $s \le 0$. The assumption that $(X, B, \mathbf{M})$ is g-lc permits $s = 0$, so

$$A(P, Y, C_Y, \mathbf{N}) = t_P = 0.$$

The centre of $P$ on $Y$ is $f(c_X(E))$, which proves the claim. □

**Lemma 24.** *Let $\pi : W \to U$ be a morphism of quasi-projective varieties, with $W$ smooth, and let $A_W$ be a semi-ample/$U$ $\mathbb{R}$-Cartier divisor. Fix a projective birational morphism $r : V \to W$ with $V$ smooth, and an snc divisor $\Theta$ on $V$ with irreducible components $E_1, \dots, E_s$. Then there is an effective $\mathbb{R}$-divisor $H \sim_{\mathbb{R},U} A_W$ such that*

$$0 \le \operatorname{mult}_E r^* H < 1 \qquad \text{for every prime divisor } E \subseteq V,$$

*while, in particular,*

$$\operatorname{mult}_{E_i} r^* H = 0 \qquad (i = 1, \dots, s),$$

*and $\Theta + r^*H$ has snc support.*

*Proof.* Since $A_W$ is semi-ample/$U$, there is a morphism $g : W \to Z$ over $U$ such that $A_W \sim_{\mathbb{R}} g^*A$, where $A$ is an ample/$U$ $\mathbb{R}$-Cartier divisor on $Z$. Let $\pi_Z : Z \to U$ be the structure morphism. Choose an ample $\mathbb{R}$-Cartier divisor $B$ on $U$ such that $A + \pi_Z^* B$ is ample. We may write

$$A + \pi_Z^* B = \sum_j c_j A_j,$$

where $c_j \in \mathbb{R}_{>0}$ and the $A_j$ are ample Cartier divisors. Choose integers $n_j > c_j$ such that $n_j A_j$ is very ample. Set

$$L_j = g^*(n_j A_j), \qquad c'_j = \frac{c_j}{n_j} \in (0, 1), \qquad A_W + \pi^* B \sim_{\mathbb{R}} \sum_j c'_j L_j.$$

Choose finite-dimensional subspaces

$$U_j \subseteq H^0(Z, \mathcal{O}_Z(n_j A_j)), \qquad U_j \otimes_k \mathcal{O}_Z \twoheadrightarrow \mathcal{O}_Z(n_j A_j).$$

The pullbacks of their generators generate $\mathcal{O}_W(L_j)$ and $\mathcal{O}_V(r^*L_j)$. Choose general members $D_j \in |U_j|$, viewed as divisors on $W$, such that $\Theta + \sum_j r^*D_j$ has snc support and

$$\operatorname{mult}_{E_i} r^*D_j = 0 \qquad \text{for all } i, j.$$

The divisors $r^*D_j$ are reduced and have no common components. Then we put

$$H = \sum_j c'_j D_j$$

is as desired; since $H \sim_{\mathbb{R}} A_W + \pi^*B$, $H \sim_{\mathbb{R},U} A_W$ and $\operatorname{mult}_E r^*H \in \{0\} \cup \{c'_j\}_j$. □

**Lemma 25.** *Let $f_W : W \to Y'$ be a surjective projective morphism over a quasi-projective variety $U$, with $W$ smooth and quasi-projective, on which $\mathbf{M}$ descends. Let $P \subseteq Y'$ be an irreducible Cartier divisor, and let $D_W$ be an $\mathbb{R}$-divisor such that $(W, D_W)$ is sub-klt over the generic point of $Y'$. Put*

$$c = t_P = \sup\big\{s \in \mathbb{R} \mid (W, D_W + sf_W^*P) \text{ is sub-lc over } \eta_P\big\}.$$

*If $A_W$ is a semi-ample/$U$ $\mathbb{R}$-Cartier divisor, there is an effective $H \sim_{\mathbb{R},U} A_W$ such that*

$$c = \sup\big\{s \in \mathbb{R} \mid (W, D_W + H + sf_W^*P) \text{ is sub-lc over } \eta_P\big\},$$

*and $(W, D_W + H)$ is still sub-klt over the generic point of $Y'$.*

*Proof.* Take a log resolution $r : V \to W$ of $(W, D_W + f_W^*P)$, and define $D_V$ by

$$K_V + D_V = r^*(K_W + D_W).$$

Lemma 16 says that $D_V + c(f_W \circ r)^*P$ has coefficients at most one over $\eta_P$, and there is a component $E$ dominating $P$ with

$$\operatorname{mult}_E\big(D_V + c(f_W \circ r)^*P\big) = 1.$$

Put $\Theta = D_V + c(f_W \circ r)^*P$. Apply the previous lemma to $r : V \to W$ and the reduced divisor supported on $\operatorname{Supp} D_V \cup \operatorname{Supp}(f_W \circ r)^*P$. We obtain an effective $H \sim_{\mathbb{R},U} A_W$ such that $\Theta + r^*H$ has snc support, every coefficient of $r^*H$ is less than one, and every component of $\operatorname{Supp} D_V \cup \operatorname{Supp}(f_W \circ r)^*P$ has multiplicity zero in $r^*H$. Therefore the snc condition gives that

$$\big(V, D_V + r^*H + c(f_W \circ r)^*P\big) \text{ is sub-lc over } \eta_P.$$

Moreover, $\operatorname{mult}_E r^*H = 0$ and $\operatorname{mult}_Z (f_W \circ r)^*P > 0$, so for $s > c$ we have

$$\operatorname{mult}_E\big(D_V + r^*H + s(f_W \circ r)^*P\big) = 1 + (s - c)\operatorname{mult}_E (f_W \circ r)^*P > 1.$$

Thus the new threshold with respect to $D_V + r^*H$ is still $c$. It is easy to see that $(W, D_W + H)$ is sub-klt over the generic point of $Y'$. □

# 3 Abundance

Let $D$ be an $\mathbb{R}$-Cartier divisor on a normal quasi-projective variety $X$ over $U$. For a prime divisor $P$ over $X$, choose a model $r : V \to X$ on which $P$ appears. The *asymptotic vanishing order* of $D$ along $P$ is

$$o_P(D) = \inf\bigl\{\operatorname{mult}_P G \mid G \in |r^*D/U|_{\mathbb{R}}\bigr\},$$

where

$$|r^*D/U|_{\mathbb{R}} = \bigl\{G \geq 0 \mid G \sim_{\mathbb{R},U} r^*D \text{ on } V\bigr\}, \qquad \inf \varnothing = +\infty.$$

Pullback and pushforward on a common higher model show that this definition is independent of the chosen model.

Assume that $X \to U$ is projective. Recall that the *invariant Iitaka dimension* of $D$ over $U$ is

$$\kappa_\iota(X/U, D) = \begin{cases} \kappa(X/U, Z), & D \sim_{\mathbb{R},U} Z \text{ for some } Z \geq 0, \\ -\infty, & \text{if no such } Z \text{ exists.} \end{cases}$$

Here the existence of $Z$ is tested after restricting to a nonempty open subset of the image of $X$ in $U$. Equivalently,

$$\kappa_\iota(X/U, D) = \kappa_\iota(X_{\bar\eta}, D|_{X_{\bar\eta}}),$$

where $X_{\bar\eta}$ is the geometric generic fibre of the contraction in the Stein factorization of $X \to U$.

If $D$ is nef/$U$, put $n = \dim X_{\bar\eta}$ and take an ample/$U$ Cartier divisor $H$ on $X$. The *numerical dimension* of $D$ over $U$ is

$$\begin{aligned} \nu(X/U, D) &= \nu(X_{\bar\eta}, D|_{X_{\bar\eta}}) \\ &= \max\bigl\{0 \leq k \leq n \mid (D|_{X_{\bar\eta}})^k \cdot (H|_{X_{\bar\eta}})^{n-k} > 0\bigr\}. \end{aligned}$$

Take a smooth model $r : V \to X$. For a Cartier divisor $A$ on $X$, put

$$\begin{aligned} \sigma(D/U; A) &= \sigma(D|_{X_{\bar\eta}}; A|_{X_{\bar\eta}}) \\ &= \max\left\{k \in \mathbb{Z}_{\geq 0} \;\middle|\; \limsup_{m\to\infty} \frac{h^0\bigl(V_{\bar\eta}, \mathcal{O}_{V_{\bar\eta}}((\lfloor mr^*D\rfloor + r^*A)|_{V_{\bar\eta}})\bigr)}{m^k} > 0\right\}, \end{aligned}$$

An empty set has value $-\infty$. The *sigma dimension* is

$$\kappa_\sigma(X/U, D) = \max_A \sigma(D/U; A),$$

where $A$ ranges over Cartier divisors on $X$. In particular, if $D$ is nef/$U$, then $\kappa_\sigma(X/U, D) = \nu(X/U, D)$.

An $\mathbb{R}$-Cartier divisor $D$ is *abundant over* $U$ if $\kappa_\iota(X/U, D) = \kappa_\sigma(X/U, D)$. If $D$ is nef/$U$, this is equivalent to $\kappa_\iota(X/U, D) = \nu(X/U, D)$. See [Hu20, Section 2.3] and [b-Div].

For a nef/$U$ $\mathbb{R}$-Cartier divisor $D$, the following conditions are equivalent.

(1) $D$ is abundant over $U$.

(2) There are a model $p : X' \to X$, a surjective morphism $f : X' \to Y$ of smooth quasi-projective varieties over $U$, and a nef and big/$U$ $\mathbb{R}$-Cartier divisor $B$ on $Y$ such that $p^*D \sim_{\mathbb{R}} f^*B$.

(3) For a general fibre $F$ of the contraction in the Stein factorization of $X \to U$, we have $o_P(D|_F) = 0$ for every prime divisor $P$ over $F$.

For the proof, see [Hu20, Lemma 2.8].

**Lemma 26.** *Let $A$ and $A'$ be $\mathbb{R}$-Cartier divisors on a normal quasi-projective variety $X$ over $U$, with $A = A' + E$, where $E \geq 0$. Suppose $|A'/U|_{\mathbb{R}} \neq \varnothing$. For a prime divisor $P$ on $X$,*

$$0 \leq o_P(A) \leq o_P(A') + \operatorname{mult}_P E.$$

*The same assertion holds for a prime divisor $P$ over $X$, using the pullback of $E$ to a model on which $P$ appears.*

*Proof.* Take $G \geq 0$ with $A' \sim_{\mathbb{R},U} G$. Then $G + E$ is effective and $G + E \sim_{\mathbb{R},U} A$, so $o_P(A) \leq \operatorname{mult}_P G + \operatorname{mult}_P E$. Taking the infimum over $G$ proves the assertion. □

**Definition 27** (Ambro model)**.** Let $f : (X, B) \to Y$ be an lc-trivial fibration over $U$, and fix an $\mathbb{R}$-Cartier divisor $L$ on $Y$ defining its moduli b-divisor $\mathbf{N}$. A model $\psi : Y' \to Y$ is an *Ambro model* for $f$ if $N_{Y'}$ is nef/$U$ and $\mathbb{R}$-Cartier and, for every higher model $r : Y'' \to Y'$, we have

$$N_{Y''} = r^* N_{Y'}.$$

In other words, $\mathbf{N}$ descends to $Y'$ with nef/$U$ trace.

**A criterion for an Ambro model.** Let $f : (X, B) \to Y$ be an lc-trivial fibration of normal quasi-projective varieties over $U$. Fix a convex decomposition

$$B = \sum_i \beta_i \Gamma_i, \qquad \beta_i > 0, \quad \sum_i \beta_i = 1,$$

where each $\Gamma_i$ is a $\mathbb{Q}$-divisor and each $f : (X, \Gamma_i) \to Y$ is a $\mathbb{Q}$-lc-trivial fibration. Consider a commutative diagram

$$\begin{array}{ccc} W & \xrightarrow{h} & X \\ {\scriptstyle f_W}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow[\psi]{} & Y, \end{array}$$

where $W$ and $Y'$ are smooth models and $f_W$ is a contraction over $U$. For each $i$, define $\Gamma_{i,W}$ by

$$K_W + \Gamma_{i,W} = h^*(K_X + \Gamma_i).$$

We require the following conditions.

(1) For each $i$, $f_W : (W, \Gamma_{i,W}) \to Y'$ is a $\mathbb{Q}$-lc-trivial fibration, and there are $\phi_i \in k(W)^\times$ and a $\mathbb{Q}$-Cartier divisor $H_i$ on $Y'$ with

$$K_W + \Gamma_{i,W} + \frac{1}{r_i} \operatorname{div}_W(\phi_i) = f_W^* H_i,$$

where $r_i$ is the least positive integer for which $r_i(K_{\overline{F}} + \Gamma_{i,W}|_{\overline{F}}) \sim 0$ on the geometric generic fibre $\overline{F}$ of $f_W$.

(2) There are reduced snc divisors $S$ on $W$ and $T$ on $Y'$, so $(W, S)$ and $(Y', T)$ are log smooth. The vertical part of $S$ is the reduced divisor supported on $f_W^{-1}(\operatorname{Supp} T)$.

(3) Put $V = Y' \setminus \operatorname{Supp} T$. The pair $(f_W^{-1}(V), S|_{f_W^{-1}(V)})$ is log smooth over $V$, i.e., for every stratum $\Theta$ of this pair, the restriction $f_W|_\Theta : \Theta \to V$ is smooth.

(4) The divisors $\Gamma_{i,W}$ and $\operatorname{div}_W(\phi_i)$ are supported on $S$. The traces on $Y'$ of the discriminant and moduli b-divisors $\mathbf{C}_i$ and $\mathbf{N}_i$ of $f_W : (W, \Gamma_{i,W}) \to Y'$, with moduli trace defined using $H_i$, are supported on $T$.

In [Hu20, Definition 3.7], this is called a resolved model of $f_W : (W, B_W) \to Y'$, where $B_W = \sum_i \beta_i \Gamma_{i,W}$. By [Hu20, Remark 3.9], $Y'$ is an Ambro model for $f_W$.

The following proposition gives a rational decomposition compatible with the discriminant and moduli b-divisors. See [Hu20, Proposition 3.6].

**Proposition 28** (Rational decomposition of an lc-trivial fibration)**.** *Let $f : (X, B) \to Y$ be an lc-trivial fibration over $U$ of normal quasi-projective varieties. Fix an $\mathbb{R}$-Cartier divisor $L$ on $Y$ with $K_X + B \sim_{\mathbb{R}} f^*L$, and let $\mathbf{C}, \mathbf{N}$ be the discriminant and moduli b-divisors defined using $L$.*

*Then there are $\mathbb{Q}$-divisors $B_i$ on $X$, $\mathbb{Q}$-Cartier divisors $L_{B_i}$ on $Y$, and real numbers $\alpha_i > 0$ with $\sum_i \alpha_i = 1$ such that*

$$B = \sum_i \alpha_i B_i, \qquad L = \sum_i \alpha_i L_{B_i}, \qquad K_X + B_i \sim_{\mathbb{Q}} f^*L_{B_i},$$

*where each $f : (X, B_i) \to Y$ is a $\mathbb{Q}$-lc-trivial fibration. If $\mathbf{C}_i, \mathbf{N}_i$ are its discriminant and moduli b-divisors defined using $L_{B_i}$, then*

$$\mathbf{C} = \sum_i \alpha_i \mathbf{C}_i, \qquad \mathbf{N} = \sum_i \alpha_i \mathbf{N}_i.$$

*If the geometric generic fibre pair $(X_{\bar\eta}, B|_{X_{\bar\eta}})$ of $f$ is klt, the pairs $(X_{\bar\eta}, B_i|_{X_{\bar\eta}})$ can also be chosen klt.*

*Proof. Step 1.* We set up the parameters for $B_i, L_i$. Write $L = \sum_k \ell_k A_k$ with $A_k$ Cartier on $Y$. Choose $\varphi_j \in k(X)^\times$ and $u_j \in \mathbb{R}$ such that

$$K_X + B + \sum_j u_j \operatorname{div}_X(\varphi_j) = \sum_k \ell_k f^* A_k.$$

For arbitrary real point $p = ((y_k)_k, (v_j)_j)$, put $L_{\Gamma(p)} = \sum_k y_k A_k$ and define

$$\Gamma(p) = f^*L_{\Gamma(p)} - K_X - \sum_j v_j \operatorname{div}_X(\varphi_j).$$

The values $y_k = \ell_k$ and $v_j = u_j$ give $(B, L)$. If $p = ((y_k)_k, (v_j)_j)$ is rational, then $\Gamma(p), L_{\Gamma(p)}$ are $\mathbb{Q}$-divisors and $K_X + \Gamma(p) \sim_{\mathbb{Q}} f^*L_{\Gamma(p)}$.

*Step 2.* Take a log resolution $h : W \to X$ of the union of the supports of $B$, the $f^*A_k$ and the $\operatorname{div}_X(\varphi_j)$. For each point $p$, the difference $\Gamma(p) - B$ is supported on these additional divisors, so $h$ also resolves every $(X, \Gamma(p))$ in Step 1. Define $B_W$ and $\Gamma_W$ by

$$K_W + B_W = h^*(K_X + B), \qquad K_W + \Gamma(p)_W = h^*(K_X + \Gamma(p)).$$

For each horizontal component $E$ of the snc divisor on $W$, including the horizontal exceptional divisors, set

$$b_E = \operatorname{mult}_E B_W, \qquad c_{p,E} = \operatorname{mult}_E \Gamma(p)_W, \qquad d_E = \max\{0, \lceil -b_E \rceil\}.$$

Let $y_k, v_j$ vary subject to $-d_E \le c_{p,E} \le 1$ for every such $E$. Then $(X, \Gamma(p))$ is sub-lc over the generic point $\eta$ of $Y$.

On the generic fibre $W_\eta$ of $f \circ h$, set

$$D_{\Gamma(p)} = \lceil -\Gamma(p)^{<1}_W \rceil|_{W_\eta}, \qquad D_B = \lceil -B^{<1}_W \rceil|_{W_\eta}.$$

The coefficient of $\lceil -\Gamma(p)^{<1}_W \rceil$ along $E$ is $\max\{0, \lceil -c_{p,E} \rceil\}$, which lies in $[0, d_E]$. Thus $0 \le D_{\Gamma(p)} \le D_B$ for such real point $p$ chosen. Taking global sections gives

$$k(Y) \subseteq H^0(W_\eta, \mathcal{O}_{W_\eta}(D_{\Gamma(p)})) \subseteq H^0(W_\eta, \mathcal{O}_{W_\eta}(D_B)). \tag{3.1}$$

By Lemma 12, over a neighborhood of $\eta$,

$$\mathcal{O}_X(\lceil \mathbf{A}^*(X, B) \rceil) = h_* \mathcal{O}_W(\lceil -B^{<1}_W \rceil).$$

Taking the generic stalk after pushing forward by $f$, the rank-one condition for $f : (X, B) \to Y$ gives $h^0(W_\eta, \mathcal{O}_{W_\eta}(D_B)) = 1$. For such real point $p$ we chosen, applying the same sheaf formula to $\Gamma(p)$ and using (3.1), we get

$$\operatorname{rank} f_* \mathcal{O}_X(\lceil \mathbf{A}^*(X, \Gamma(p)) \rceil) = 1.$$

Hence $f : (X, \Gamma(p)) \to Y$ is lc-trivial, and is $\mathbb{Q}$-lc-trivial when all the parameters in $q$ are rational.

Denote $p_0 = ((\ell_k)_k, (u_j)_j)$. Take a large enough positive integer $M$ and set

$$\mathcal{P} = \left\{ p = ((y_k)_k, (v_j)_j) \;\middle|\; \begin{array}{l} -d_E \le c_{p,E} \le 1 \text{ for every horizontal component } E, \\ |y_k| \le M, \quad |v_j| \le M \text{ for every } k, j \end{array} \right\}.$$

If $(X_{\bar\eta}, B|_{X_{\bar\eta}})$ is klt, each $b_E < 1$. Choose rational numbers $q_E$ with $b_E < q_E < 1$. Impose the additional conditions

$$\mathcal{P}_{\mathrm{klt}} = \left\{ p \in \mathcal{P} \;\middle|\; c_{p,E} \le q_E \text{ for every such } E, \quad \Gamma|_{X_\eta} \ge 0 \right\}.$$

The last condition consists of non-negativity inequalities on the coefficients of the horizontal components of $\Gamma(p)$ on $X$. These are rational polytopes containing $p_0$ in their respective cases. Choose a convex decomposition in $\mathcal{P}$ or $\mathcal{P}_{\mathrm{klt}}$ as appropriate,

$$p_0 = \sum_{i=1}^{v} a_i p_i, \qquad a_i > 0, \qquad \sum_{i=1}^{v} a_i = 1,$$

where the $p_i$ are rational vertices. Each $p_i$ gives $(\Gamma_i = \Gamma(p_i), L_i = L_{\Gamma_i})$ defining a $\mathbb{Q}$-lc-trivial fibration $f : (X, \Gamma_i) \to Y$, with

$$K_X + \Gamma_i \sim_{\mathbb{Q}} f^* L_i, \qquad B = \sum_i a_i \Gamma_i, \qquad L = \sum_i a_i L_i.$$

*Step 3.* We construct $W, Y', S, T$ as in the resolved model above, with $W$ dominating the log resolution in Step 2. For each $i$, we write $K_W + \Gamma_{i,W} = h^*(K_X + \Gamma_i)$. Since $K_X + \Gamma_i \sim_{\mathbb{Q}} f^* L_i$, there is a least positive integer $r_i$ such that

$$r_i(K_{\overline{F}} + \Gamma_i|_{\overline{F}}) \sim 0$$

on the geometric generic fibre $\overline{F}$ of $f$. The index is the same on the generic fibre by flat base change for global sections of a line bundle. We can therefore choose $\phi_i \in k(X)^\times$ with $\operatorname{div}_{X_\eta}(\phi_i|_{X_\eta}) = -r_i(K_{X_\eta} + \Gamma_i|_{X_\eta})$.

Write

$$K_X + \Gamma_i + \frac{1}{m_i}\operatorname{div}_X(\zeta_i) = f^*L_i, \qquad r_i \mid m_i,$$

with $\zeta_i \in k(X)^\times$. The ratio $\zeta_i/\phi_i^{m_i/r_i}$ has zero divisor on the generic fibre. It equals $f^*\theta_i$ for some $\theta_i \in k(Y)^\times$.[2] Put $H_{i,Y} = L_i - \frac{1}{m_i}\operatorname{div}_Y(\theta_i)$. Then

$$K_X + \Gamma_i + \frac{1}{r_i}\operatorname{div}_X(\phi_i) = f^*H_{i,Y}. \tag{3.2}$$

Here $H_{i,Y}$ is $\mathbb{Q}$-Cartier.

Take a log resolution $h_0 : W_0 \to X$ dominating the resolution in Step 2. Write $\Gamma_{i,W_0}$ for the crepant boundary on $W_0$ and take a reduced snc divisor $S_0$ containing

$$\bigcup_k \operatorname{Supp} h_0^* f^* A_k \cup \bigcup_i \Big( \operatorname{Supp}\operatorname{div}_{W_0}(\phi_i) \cup \operatorname{Supp}\operatorname{div}_{W_0}(\zeta_i)\Big) \cup \bigcup_j \operatorname{Supp}\operatorname{div}_{W_0}(\varphi_j) \cup \operatorname{Supp} B_{W_0},$$

while note that $\Gamma(p)_{W_0} = B_{W_0} + \sum_k (y_k - \ell_k) h_0^* f^* A_k - \sum_j (v_j - u_j)\operatorname{div}_{W_0}(\varphi_j)$ for each $p$. Denote $g = f \circ h_0$. Remove from $Y$ the images $g(D)$ of all vertical strata $D$ of $(W_0, S_0)$. For every horizontal stratum $D$, use generic smoothness to make $g|_D$ smooth over the remaining open subset. We also require this open subset to lie in the smooth locus of $Y$ and $g$ to be smooth over it. Furthermore, remove the supports of $C(\Gamma_i)_Y$ and $H_{i,Y} - K_Y - C(\Gamma_i)_Y$ for all $i$. Denote the resulting open subset by $V_0$. Here $C(\Gamma_i)_Y$ is the discriminant trace for $f : (X, \Gamma_i) \to Y$, and its moduli trace $N(\Gamma_i)_Y$ is defined using $H_{i,Y}$.

Resolve the ideal defining $Y \setminus V_0$ to obtain a smooth model $\psi : Y' \to Y$ such that $\psi|_V : V \cong V_0$, where $V = \psi^{-1}(V_0)$, and $\psi^{-1}(Y \setminus V_0)$ has snc support. Put

$$T = (\psi^{-1}(Y \setminus V_0))_{\mathrm{red}}, \qquad H_i = \psi^* H_{i,Y}.$$

The isomorphism $\psi|_V : V \cong V_0$ gives

$$C(\Gamma_i)_{Y'}|_V = (\psi|_V)^*(C(\Gamma_i)_Y|_{V_0}) = 0,$$

and

$$\begin{aligned} N(\Gamma_i)_{Y'}|_V &= (H_i - K_{Y'} - C(\Gamma_i)_Y|_V \\ &= (\psi|_V)^*(H_{i,Y} - K_Y - C(\Gamma_i)_Y|_{V_0}) \\ &= (\psi|_V)^*(N(\Gamma_i)_Y|_{V_0}) = 0. \end{aligned}$$

Thus the traces of two b-divisors corresponding to $\Gamma_i$ are supported on $T$ for each $i$.

Finally, take the main component of $W_0 \times_Y Y'$ and resolve the inverse images of $\operatorname{Supp} T$ and $\operatorname{Supp} S_0$, leaving the inverse image of $V$ unchanged. This gives a smooth model $W$ with a morphism $f_W : W \to Y'$. The map $W \to W_0$ induces an isomorphism $f_W^{-1}(V) \cong g^{-1}(V_0)$. Put

$$S = (f_W^{-1}(\operatorname{Supp} T))_{\mathrm{red}} + \sum_{\substack{D \text{ a horizontal} \\ \text{component of } S_0}} \widetilde{D},$$

[2] We may regard $F$ as a very general fibre of $f$, since they are isomorphic as abstract schemes. Then $\operatorname{div}_F \zeta_i/\phi_i^{m_i/r_i} = 0$ implies that the components of $\operatorname{div}_X \zeta_i/\phi_i^{m_i/r_i}$ won't dominates $Y$, they are vertical. Choose an open subset $U \subseteq Y$ which avoids the image of poles of vertical components, we can get an element in $\mathcal{O}_X(f^{-1}U)$. Since $f$ is a contraction, we have $f^* : \mathcal{O}_Y(U) \to \mathcal{O}_X(f^{-1}U)$, so $\zeta_i/\phi_i^{m_i/r_i} = f^*\theta_i$ for some $\theta_i \in k(Y)^\times$.

where $\widetilde{D}$ is the strict transform of $D$ under $W \to W_0$. Pulling back (3.2) along $h : W \to X$ gives

$$K_W + \Gamma_{i,W} + \frac{1}{r_i}\operatorname{div}_W(\phi_i) = f_W^* H_i.$$

This is the desired diagram satisfying the four conditions of the resolved model.
*Step 4.* Write $t_P(\Gamma(p))$ to be the sub-lc threshold of $(X, \Gamma(p))$ at a prime $P$ over $Y$, that is

$$t_P(\Gamma(p)) = \sup\big\{s \in \mathbb{R} \mid (W, \Gamma(p)_W + sf_W^* P) \text{ is sub-lc over } \eta_P\big\}.$$

The crepant boundary $\Gamma(p)_W$ is supported in $S$ since it is the linear combination of $h^* f^* A_k$, $\operatorname{div}_W(\varphi_j)$ and $B_W$.

First let $P$ be a prime divisor on $Y'$ which is not a component of $T$, such that $\eta_P \in V$. For this calculation, we possibly shrink $V$ near $\eta_P$ such that $P_V = P \cap V$ is non-singular. Take the base change of two smooth morphisms

$$f_W^{-1}(V) \longrightarrow V, \qquad D \cap f_W^{-1}(V) \longrightarrow V$$

along the inclusion $P_V \to V$, where $D$ is an arbitrary stratum of $S$, we can obtain that

$$\begin{aligned} f_W^{-1}(V) \times_V P_V = f_W^{-1}(P_V) &\longrightarrow P_V, \\ \big(D \cap f_W^{-1}(V)\big) \times_V P_V = D \cap f_W^{-1}(P_V) &\longrightarrow P_V, \end{aligned}$$

are smooth morphisms. Therefore, $f_W^{-1}(P_V)$ and its intersection with any stratum of $S$ are smooth over the base field. The restriction of each stratum of $S$ on $f_W^{-1}(P_V)$ is flat over $P_V$, so pulling back the local equation of $P_V$ cuts out a Cartier divisor on the stratum of $S$ wherever they meet $f_W^{-1}(P_V)$. Thus $f_W^* P$ is reduced over $\eta_P$ and meets every stratum of $S$ in expected codimension, which gives snc support over $\eta_P$ after adding $S$.

Since all components of $\Gamma(p)_W$ over $V$ will dominate $Y'$ for every vertex $p_i$, by our construction, the sub-pair $(X, \Gamma(p))$ being sub-lc over the generic point of $Y$ implies that

$$\operatorname{mult}_E \Gamma(p)_W \leq 1$$

where $E$ is the component of $\Gamma(p)_W$ meeting $f_W^{-1}(V)$. If $\operatorname{mult}_E f_W^* P \neq 0$ over $\eta_P$, i.e. $E$ is a component of $f_W^* P$ which dominates $P$, then $\operatorname{mult}_E \Gamma(p)_W = 0$; otherwise $E$ is a vertical component of $S$ by $\operatorname{Supp} \Gamma(p)_W \subseteq \operatorname{Supp} S$, and hence $P = f_W(E) \subseteq \operatorname{Supp} T$ which contradicts the fact $\eta_P \in V = Y' \setminus \operatorname{Supp} T$. Therefore,

$$\operatorname{mult}_E\big(\Gamma(p)_W + sf_W^* P\big) = 0 + s \cdot 1 = s,$$

where $\operatorname{mult}_E f_W^* P = 1$ by $f_W^* P$ is reduced over $\eta_P$. The snc condition implies that $(W, \Gamma(p)_W + sf_W^* P)$ is sub-lc over $\eta_P$ if and only if its coefficients are less than one. This proves that $t_P(\Gamma(p)) = 1$ for every $p \in \mathcal{P}$ and every $P$ meets $V$.

Now, let $P$ be a component of $T$, we put

$$m_E = \operatorname{mult}_E f_W^* P, \qquad \lambda_{p,E} = \operatorname{mult}_E \Gamma(p)_W.$$

The threshold formula, Lemma 16 gives that

$$\ell_{P,E}(p) = \frac{1 - \lambda_{p,E}}{m_E}, \qquad t_P = \min_E \ell_{P,E}.$$

We denote by $E_P$ the component of $B_W + t_P(B) f_W^* P$ such that

$$\operatorname{mult}_{E_P}\big(B_W + t_P(B) f_W^* P\big) = 1.$$

Consider the rational polytope

$$\mathcal{P}_P = \mathcal{P} \cap \bigcap_E \big\{p \mid \ell_{P,E_P}(p) - \ell_{P,E}(p) \le 0\big\} = \big\{p \in \mathcal{P} \mid \ell_{P,E_P}(p) \le \ell_{P,E}(p)\big\},$$

where $E$ varies the components of $f_W^* P$ dominating $P$. Note that we already have $p_0 \in \mathcal{P}_P$, and for each $p \in \mathcal{P}_P$ we have

$$t_P(\Gamma(p)) = \min_E \ell_{P,E}(p) = \ell_{P,E_P}(p).$$

Take rational vertices $q_i$ of $\mathcal{P}_P$ and write $p_0 = \sum_i \alpha_i q_i$, where $\sum \alpha_i = 1$. Define $B_i = \Gamma(q_i)$, then we have

$$B = \sum_i \alpha_i B_i, \qquad L = \sum_i \alpha_i L_{B_i}, \qquad t_P(B) = \sum_i \alpha_i t_P(B_i),$$

while the last expression gives that

$$1 - t_P(B) = \sum_i \alpha_i \big(1 - t_P(B_i)\big).$$

Each $K_X + B_i \sim_{\mathbb{Q}} f^* L_{B_i}$ and Step 2 defines a $\mathbb{Q}$-lc-trivial $f : (X, B_i) \to Y$, and hence we have

$$C_{Y'} = \sum_i \alpha_i C_{i,Y'}, \qquad N_{Y'} = \sum_i \alpha_i N_{i,Y'}.$$

*Step 5.*

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